

COM-480 Data Visualization: Milestone 2

Thirst for Data: China's Beverage Market Analysis

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1. Project Goal

Our project aims to provide a comprehensive analysis of the beverage market landscape in China, focusing on brand competition, consumer behavior, and regional differences. By integrating a **Shops Dataset** (1,219 brands) with a granular **Consumer Dataset** (31,800 orders), we seek to bridge the gap between high-level market trends and personalized user experiences. The visualization will help users understand how pricing, scale, and demographics interact in one of the world's most dynamic consumer markets.

2. Core Visualization (Minimal Viable Product)

To fulfill the core requirements of the project, we will implement the following "Brand Market Landscape" components as our MVP:

- **Interactive Bubble/Logo Cloud:** A central visualization showing the Top 20 drink shops. Bubble radius will represent the *Store Count*, allowing users to identify market leaders instantly.
- **Brand Matrix (Scatter Plot):** A "Price vs. Store Count" plot to categorize brands into premium, mass-market, and niche segments.
- **Faceted Price Histograms:** Distribution charts segmented by brand type (Tea, Coffee, Milk drinks) to show pricing strategies across different sectors.
- **Functional Website Skeleton:** A structured multi-page layout with a navigation bar to switch between market overview and behavior analysis.

3. Extra Ideas & Enhanced Features

These modular features will be added to enhance interactivity and storytelling depth:

- **Consumer Behavior Dashboard:** Four synchronized charts (Pie/Donut) visualizing:
 - *Scenario & Motive:* Delivery vs. dine-in; Social vs. Novelty seeking.
 - *Demographics:* Distributions across Gender and Age-groups.
- **Regional Tier Analysis (Pyramid):** A hierarchical visualization representing 1st, 2nd, and 3rd-tier cities to show brand penetration.
- **3rd-Tier City Drill-down Map:** A detailed geospatial view focusing on emerging markets (e.g., Xiangyang, Shantou) from 2019 to 2023.
- **Personalized Recommendation Tool:** A decision-tree style interactive widget that suggests brands based on user-inputted demographics.

